1st Midterm · Unit 3 · Constant coefficients, Cauchy-Euler and forcing · 5 points · 30 min · High difficulty / trap

P04 — Inverse problem: rebuild $y'' + a\,y' + b\,y = f$ from one homogeneous solution and one particular solution, then the resonant forcing e^{-x} sin 2x, an IVP and which forcings need the modified guess

The real constants a, b of the equation

$$y'' + a\,y' + b\,y = f(x)$$

are unknown. Two facts are known about it:

- *the function $y = e^{-x}\cos 2x$ solves the homogeneous equation ($f = 0$);*
- *when the right-hand side is $f(x) = k\,x$, for a certain constant k, the polynomial $y = 5x - 2$ is a solution.*

Afterwards the equation is driven by $f(x) = 8\,e^{-x}\sin 2x$, with initial data $y(0) = 1$, $y'(0) = -3$. Finally, consider forcings of the form $f(x) = e^{\alpha x}\cos\omega x$ or $e^{\alpha x}\sin\omega x$ with real α and $\omega \geq 0$, and in particular the purely oscillatory forcing $f(x) = \cos\omega x$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns: first the parameters a, b, k (inverse problem), then the function $y(x)$, $x \in \mathbb{R}$.
- Operator: second order, linear, constant coefficients, $L[y] = y'' + ay' + by$. Coefficients and forcings are continuous on $\mathbb{R}$, so every IVP has a unique solution on all of $\mathbb{R}$.
- Superposition: $y = y_h + y_p$, with y_h from the characteristic equation $r^2 + ar + b = 0$ and y_p from undetermined coefficients.

Phase 2. Canonical formulation

The equation is already in standard form. Writing $y = e^{-x}u$ (an exponential shift, **extension**: not in the course, it only organises the computation of (b)) turns L into an operator on u that will expose the resonance.

Part (a) · 1.5 points

Reconstruction of the equation

Find a, b and k, and write the general solution of the homogeneous equation.

Step 1. Derivatives of the given solution

Product and chain rule on $y = e^{-x}\cos 2x$:

$$y' = -e^{-x}\cos 2x - 2e^{-x}\sin 2x = e^{-x}\big(-\cos 2x - 2\sin 2x\big),$$

$$y'' = -e^{-x}\big(-\cos 2x - 2\sin 2x\big) + e^{-x}\big(2\sin 2x - 4\cos 2x\big) = e^{-x}\big(-3\cos 2x + 4\sin 2x\big).$$

Step 2. Conditions on a and b

$$L[y] = e^{-x}\Big[\big(-3 - a + b\big)\cos 2x + \big(4 - 2a\big)\sin 2x\Big] = 0 \quad \forall x.$$

$e^{-x} \neq 0$ and $\cos 2x$, $\sin 2x$ are linearly independent, so both brackets vanish: $4 - 2a = 0 \Rightarrow a = 2$, and $-3 - 2 + b = 0 \Rightarrow b = 5$.

Cross-check: $e^{-x}\cos 2x = \operatorname{Re} e^{(-1+2i)x}$ requires the root $r = -1 + 2i$; real coefficients give also $-1 - 2i$, so $r^2 + ar + b = (r + 1 - 2i)(r + 1 + 2i) = (r+1)^2 + 4 = r^2 + 2r + 5$.

Step 3. The constant k

$y = 5x - 2$: $y' = 5$, $y'' = 0$, so $L[y] = 0 + 2\cdot 5 + 5(5x - 2) = 25x$. Hence $k = 25$. Undetermined coefficients give the same: with $y_p = Ax + B$, $L[y_p] = 2A + 5Ax + 5B = 25x$, so $5A = 25$, $A = 5$, and $2A + 5B = 0$, $B = -2$. The second fact is consistent with $a = 2$, $b = 5$ (it imposes $5a = 2b$ through the constant term, which they satisfy).

Step 4. Homogeneous solution

$r^2 + 2r + 5 = 0$: $r = \dfrac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i$ (Robinson Ch. 12), so

$$y_h = e^{-x}\big(c_1\cos 2x + c_2\sin 2x\big).$$

KEY RESULTS

- $a = 2, b = 5$; characteristic roots $r = -1 \pm 2i$.
- $k = 25$; $y_p = 5x - 2$ is recovered by the guess $Ax + B$.
- $y_h = e^{-x}\big(c_1 \cos 2x + c_2 \sin 2x\big)$.

Part (b) · 1.5 points

Resonant forcing and the modified guess

For $f(x) = 8\,e^{-x} \sin 2x$ explain why the usual guess fails and find a particular solution.

Step 1. Why the usual guess fails

The forcing is $\operatorname{Im}\left(8e^{(-1+2i)x}\right)$ and $-1 + 2i$ is a simple root of $r^2 + 2r + 5$. The guess $e^{-x}(A\cos 2x + B\sin 2x)$ lies in y_h, so L of it is 0 for every A, B, never $8e^{-x}\sin 2x$. The rule (Robinson 14.5) multiplies the guess by x (once, because the root is simple).

Step 2. Exponential shift

Let $y = e^{-x}u$. By the product rule, $y' = e^{-x}(u' - u)$ and $y'' = -e^{-x}(u' - u) + e^{-x}(u'' - u') = e^{-x}(u'' - 2u' + u)$. Then

$$L[y] = e^{-x}\Big[(u'' - 2u' + u) + 2(u' - u) + 5u\Big] = e^{-x}\big(u'' + 4u\big).$$

So $L[y] = 8e^{-x}\sin 2x$ becomes $u'' + 4u = 8\sin 2x$: an undamped oscillator forced exactly at its natural frequency 2.

Step 3. Coefficients

With $u = x(A\cos 2x + B\sin 2x)$:

$$u' = (A\cos 2x + B\sin 2x) + x(-2A\sin 2x + 2B\cos 2x),$$

$$u'' = 2(-2A\sin 2x + 2B\cos 2x) + x(-4A\cos 2x - 4B\sin 2x).$$

The terms with x cancel against $4u$: $x(-4A\cos 2x - 4B\sin 2x) + 4x(A\cos 2x + B\sin 2x) = 0$, so $u'' + 4u = 4B\cos 2x - 4A\sin 2x$. Matching with $8\sin 2x$: $4B = 0$ and $-4A = 8$, so $A = -2$, $B = 0$:

$$y_p = -2x\,e^{-x}\cos 2x.$$

KEY RESULTS

- $e^{-x}(A\cos 2x + B\sin 2x)$ is a homogeneous solution: L annihilates it.
- Modified guess $y_p = x\,e^{-x}(A\cos 2x + B\sin 2x)$; $L[y_p] = e^{-x}\big(4B\cos 2x - 4A\sin 2x\big)$.
- $A = -2, B = 0$: $y_p = -2x\,e^{-x}\cos 2x$.

Part (c) · 1 point

Initial value problem

Solve the IVP with $f(x) = 8e^{-x}\sin 2x$, $y(0) = 1$, $y'(0) = -3$, and describe its behaviour as $x \to \infty$.

Step 1. General solution and its derivative

$y = e^{-x}(c_1\cos 2x + c_2\sin 2x) - 2xe^{-x}\cos 2x$. Differentiating with the product rule and evaluating at $x = 0$ ($\cos 0 = 1, \sin 0 = 0$):

$$\frac{d}{dx}\Big[e^{-x}(c_1\cos 2x + c_2\sin 2x)\Big]_{x=0} = -c_1 + 2c_2, \qquad \frac{d}{dx}\Big[-2xe^{-x}\cos 2x\Big]_{x=0} = -2e^{0}\cos 0 = -2.$$

Step 2. Constants

$y(0) = c_1 = 1$; $y'(0) = -c_1 + 2c_2 - 2 = -3 \Rightarrow 2c_2 = 0, c_2 = 0$:

$$y(x) = (1 - 2x)\,e^{-x}\cos 2x.$$

Step 3. Long-time behaviour

$|y| \le |1 - 2x|e^{-x} \to 0$, because e^{-x} beats any polynomial. Here "resonance" does not mean unbounded growth: the factor x only slows the decay, since the damping ($a = 2 > 0$) sets the envelope e^{-x}.

KEY RESULTS

- $y'(0) = -c_1 + 2c_2 - 2$; $c_1 = 1, c_2 = 0$.
- $y(x) = (1 - 2x)\,e^{-x}\cos 2x \to 0$ as $x \to \infty$.

Part (d) · 1 point

Which forcings need the modified guess

For $f = e^{\alpha x}\cos\omega x$ or $e^{\alpha x}\sin\omega x$, when is the modified guess needed? For $f = \cos\omega x$, find the amplitude of the particular solution as a function of ω and the frequency that maximises it.

Step 1. The rule

The guess $e^{\alpha x}(A\cos\omega x + B\sin\omega x)$ fails only when it solves the homogeneous equation, i.e. when $\alpha + i\omega \in \{-1+2i, -1-2i\}$: $\alpha = -1$ and $\omega = 2$. For a polynomial or $e^{\alpha x}$ ($\omega = 0$) the guess would need a real root, and there is none. For $\cos\omega x$ ($\alpha = 0$) the real part 0 never equals -1: **no frequency** of a purely oscillatory forcing needs the modified guess.

Step 2. Coefficients for cos(omega x)

$y_p' = -A\omega\sin\omega x + B\omega\cos\omega x$, $y_p'' = -\omega^2 y_p$. Then

$$L[y_p] = \left[(5-\omega^2)A + 2\omega B\right]\cos\omega x + \left[(5-\omega^2)B - 2\omega A\right]\sin\omega x = \cos\omega x.$$

From the second bracket $B = \frac{2\omega}{5-\omega^2}A$ (for $\omega^2 \neq 5$); substituting into the first, $A\,\frac{(5-\omega^2)^2+4\omega^2}{5-\omega^2} = 1$. Hence $A = \frac{5-\omega^2}{D}$, $B = \frac{2\omega}{D}$ with $D = (5-\omega^2)^2 + 4\omega^2$ (the same formulas hold at $\omega^2 = 5$, solving directly).

Step 3. Amplitude and its maximum

$R = \sqrt{A^2+B^2} = \dfrac{\sqrt{(5-\omega^2)^2+4\omega^2}}{D} = \dfrac{1}{\sqrt{D}}$. Expanding, $D = \omega^4 - 10\omega^2 + 25 + 4\omega^2 = \omega^4 - 6\omega^2 + 25 = (\omega^2-3)^2 + 16 \geq 16$. D is smallest, and R largest, at $\omega^2 = 3$:

$$\omega = \sqrt{3}, \qquad R_{\max} = \frac{1}{\sqrt{16}} = \frac{1}{4}.$$

This "practical resonance" is a finite peak, not a failure of the guess: it is the damped counterpart of the undamped resonance, and it sits below the frequency 2 of the free oscillation.

KEY RESULTS

- Modified guess only for $\alpha = -1, \omega = 2$. Never for $\cos\omega x$ ($\alpha = 0$), for polynomials or for real exponentials.
- $A = \dfrac{5-\omega^2}{D}, B = \dfrac{2\omega}{D}, D = (5-\omega^2)^2 + 4\omega^2 = (\omega^2-3)^2 + 16$; amplitude $R(\omega) = \dfrac{1}{\sqrt{D}}$.
- Maximum at $\omega = \sqrt{3}$, with $R_{\max} = \dfrac{1}{4}$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** as $x \to \infty$ every solution of the homogeneous equation decays like e^{-x}, so the long-time response to $\cos \omega x$ is y_p alone; $R(0) = \frac{1}{5}$ (static response $1/b$) and $R(\omega) \sim \omega^{-2}$ as $\omega \to \infty$.
- **Checks:** $L[e^{-x}\cos 2x] = 0$, $L[5x - 2] = 25x$, $L[-2xe^{-x}\cos 2x] = 8e^{-x}\sin 2x$ and the IVP were verified by substitution. With zero damping and forcing at the natural frequency (for example $y'' + 4y = 8\sin 2x$) the factor x would grow without bound (true resonance); here $a = 2$ gives the envelope e^{-x}, so $xe^{-x} \to 0$.
- **Common mistakes in the exam:** (1) deducing only b from the homogeneous solution and forgetting that the $\sin 2x$ coefficient fixes a; (2) trying $e^{-x}(A\cos 2x + B\sin 2x)$ for the resonant forcing and getting $0 = 8$; (3) calling $\omega = 2$ or $\omega = \sqrt{3}$ resonant for $\cos \omega x$: the guess never needs modifying there; (4) losing the term -2 from $\frac{d}{dx}(-2xe^{-x}\cos 2x)$ at $x = 0$.